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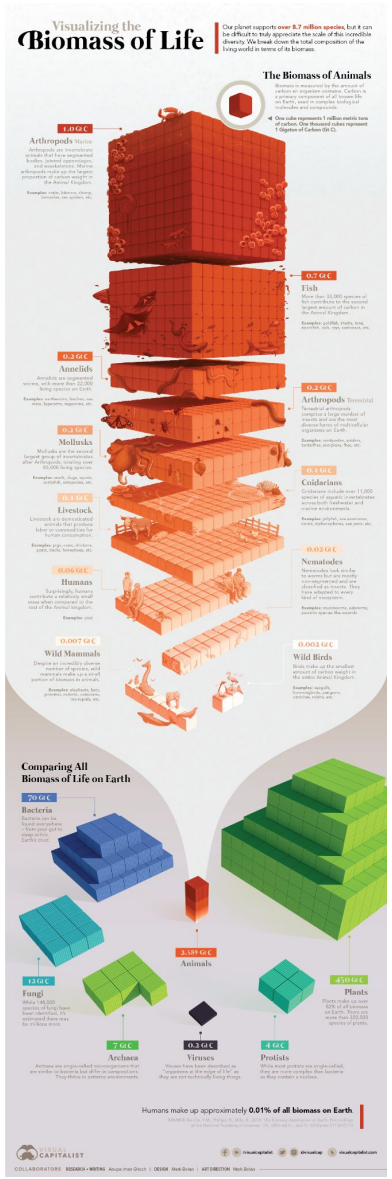
FIT2179
Data Visualisation

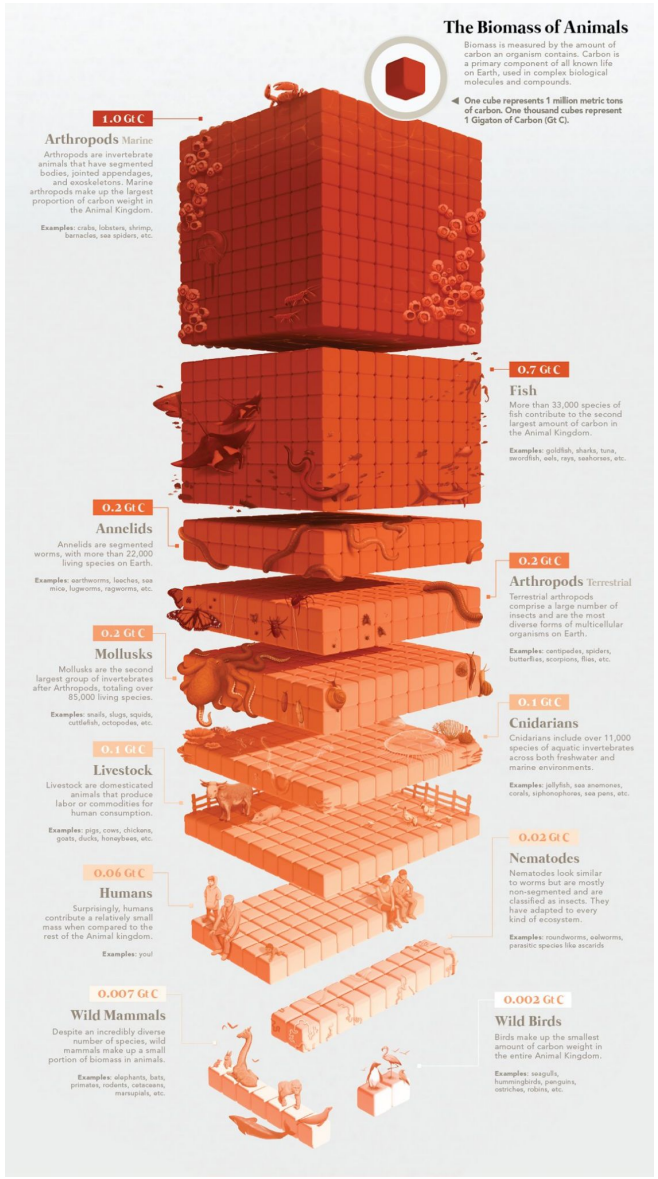
Week 5 Workshop

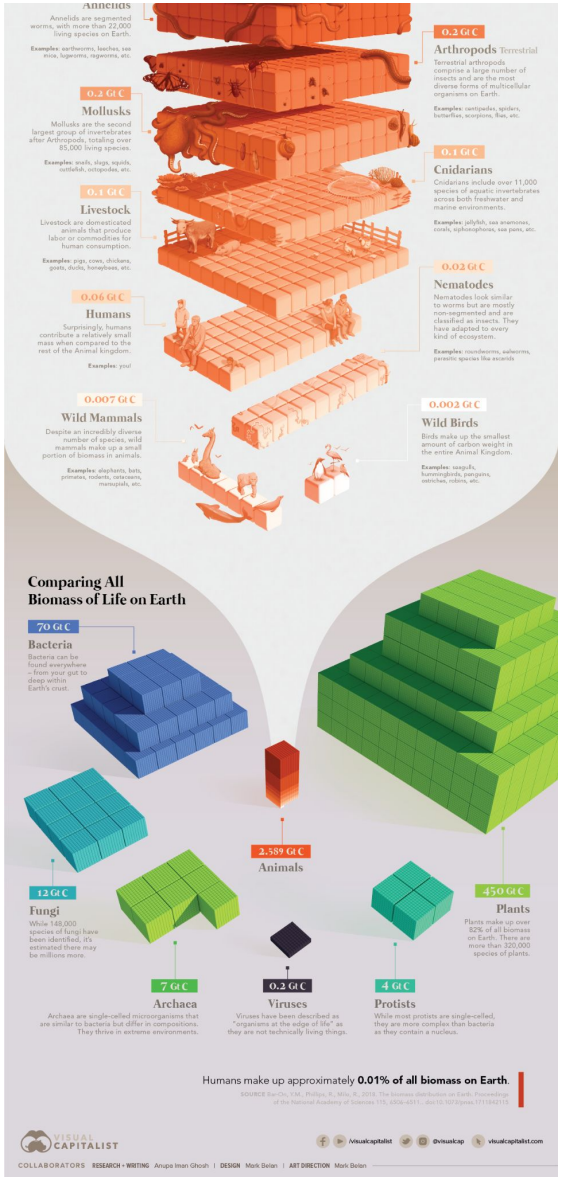
Visual Idioms for Tables

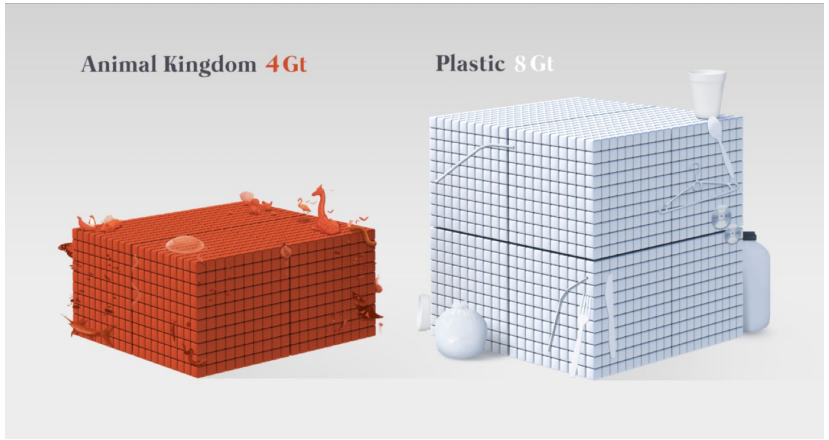


Visualisation of the Week





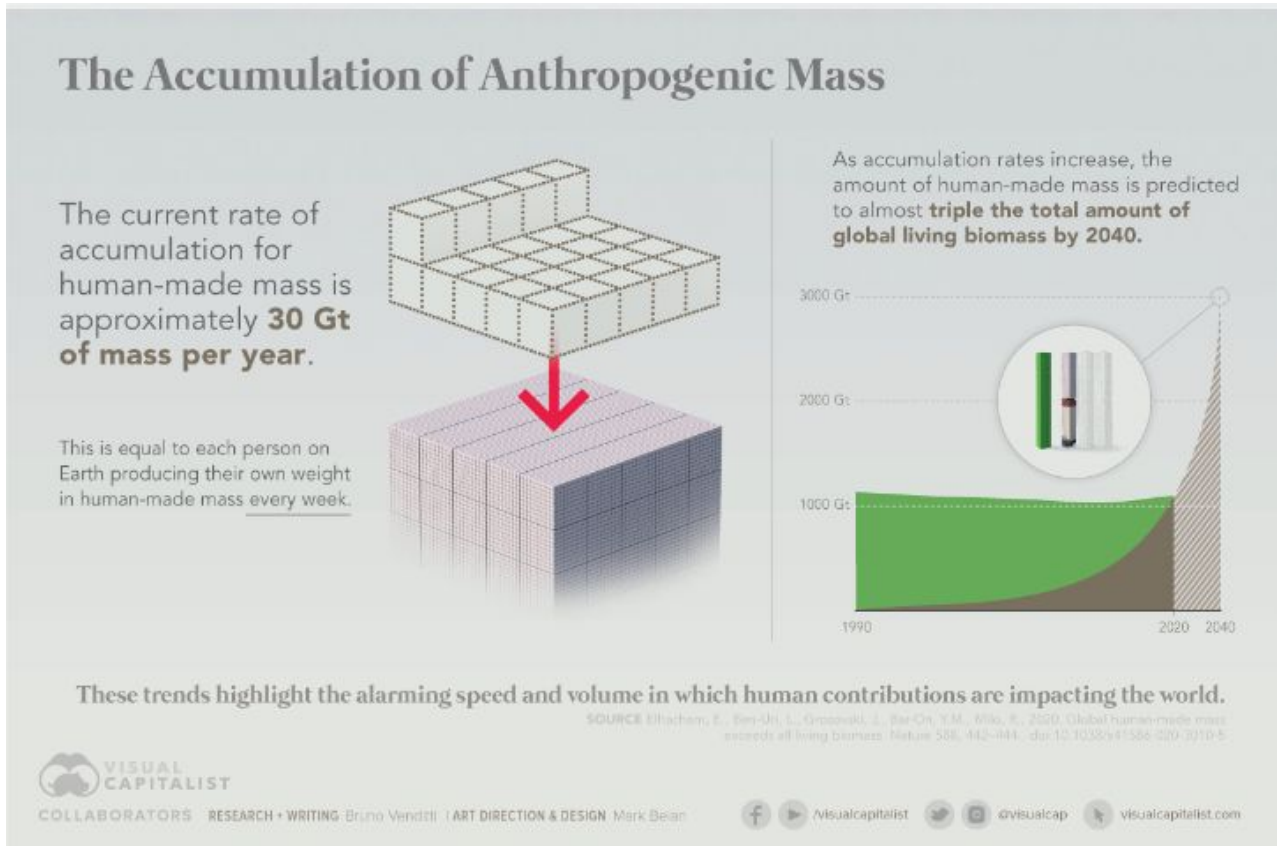


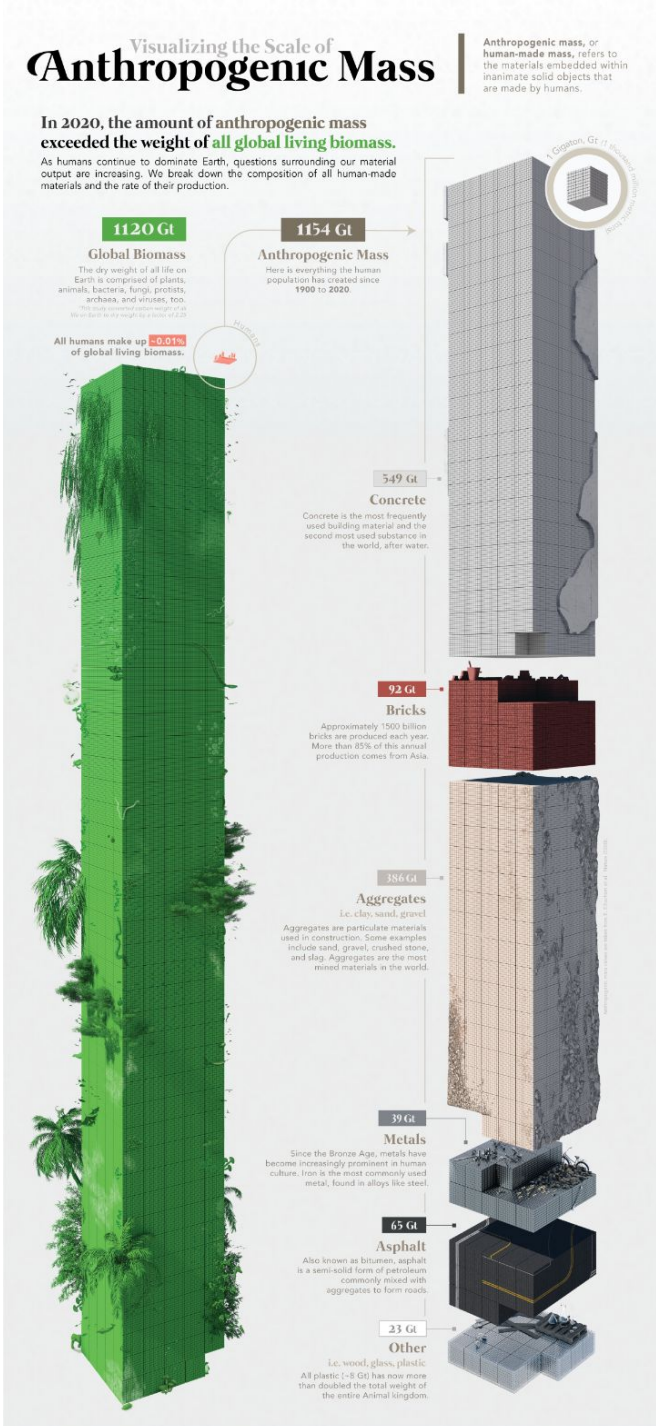


https://www.visualcapitalist.com/visualizing-the-accumulation-of-human-made-mass-on-earth/

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Visualisation of the Week





https://www.visualcapitalist.com/visualizing-the-accumulation-of-human-made-mass-on-earth/

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Weeks 5 and 6

1. **Data Visualisation 1:** You should have
    1. … found datasets
    2. … completed a first draft in Tableau
2. **Workshop Week 5, now:** Test 1 (assessed, 10%)
3. **Studio Week 5**
    1. Discussion and feedback Week 4 posting in studio or on forum
    2. Introduction to CSS
4. **Studio Week 6**: Project Critique (assessed, 3%)
5. **Workshop Week 6:** Networks and tree (assessed, 3%)

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Any question about videos or readings?

1. Visualisation idioms for table datasets
    1. Venn diagram
    2. Proportional symbol chart (packed bubble chart), word cloud
    3. Matrix chart, waffle chart
    4. ISOTYPE and effect of icons
    5. Histogram
    6. Heat map
    7. Box plot
    8. Slope chart
    9. Parallel coordinate plot
    10. Bump chart
    11. Streamgraph
    12. Radar chart
    13. Spiral plot
2. Repeating patterns
    1. Periodic line charts, overlaying
    2. Aggregation > heat map
    3. Small multiples
    4. Animation

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Tableau Exercise Week 5

Tableau Exercise

FIT2179 Week 5 Tableau Exercise ZIP

This exercise is not marked, but strongly recommended. Post your questions about this exercise on the Ed forum.

2.4 MB

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Workshop Activities

1. Identifying idioms
2. Ranking idioms
3. Designing Isotype visualisations

Reminder: Catalogues for choosing visualisation idioms

- https://flowingdata.com/chart-types/
- https://www.data-to-viz.com
- http://datavizproject.com
- http://www.datavizcatalogue.com
- Visual Vocabulary
- Tableau Chart Catalogue: highly recommended step-by-step instruction for building more advanced visualisation idioms with Tableau.

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Activity 1 – Identifying Idioms

For two visualisations, use the online catalogues listed below:

- Identify the visual idioms in the examples
- For each idiom, describe
  - What? - What data is it good for?
  - How? - How does it visualise data (i.e. marks and channels)?
  - Are there similar idioms?
  - Pros and cons?

Catalogues

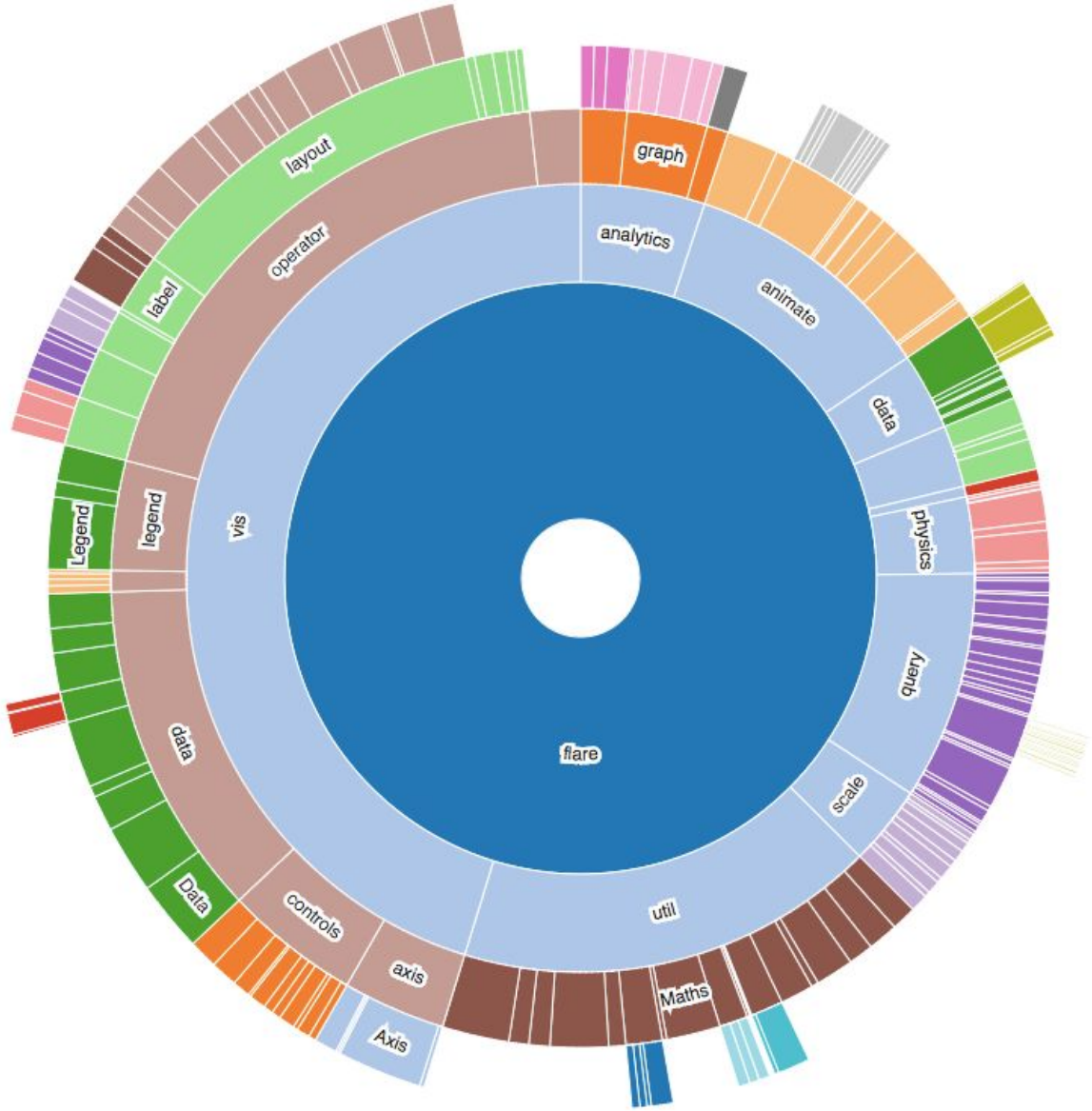
- FT Visual Vocabulary (also Visual Vocabulary on Github)
- The Data Visualisation Catalogue
- From data to Viz

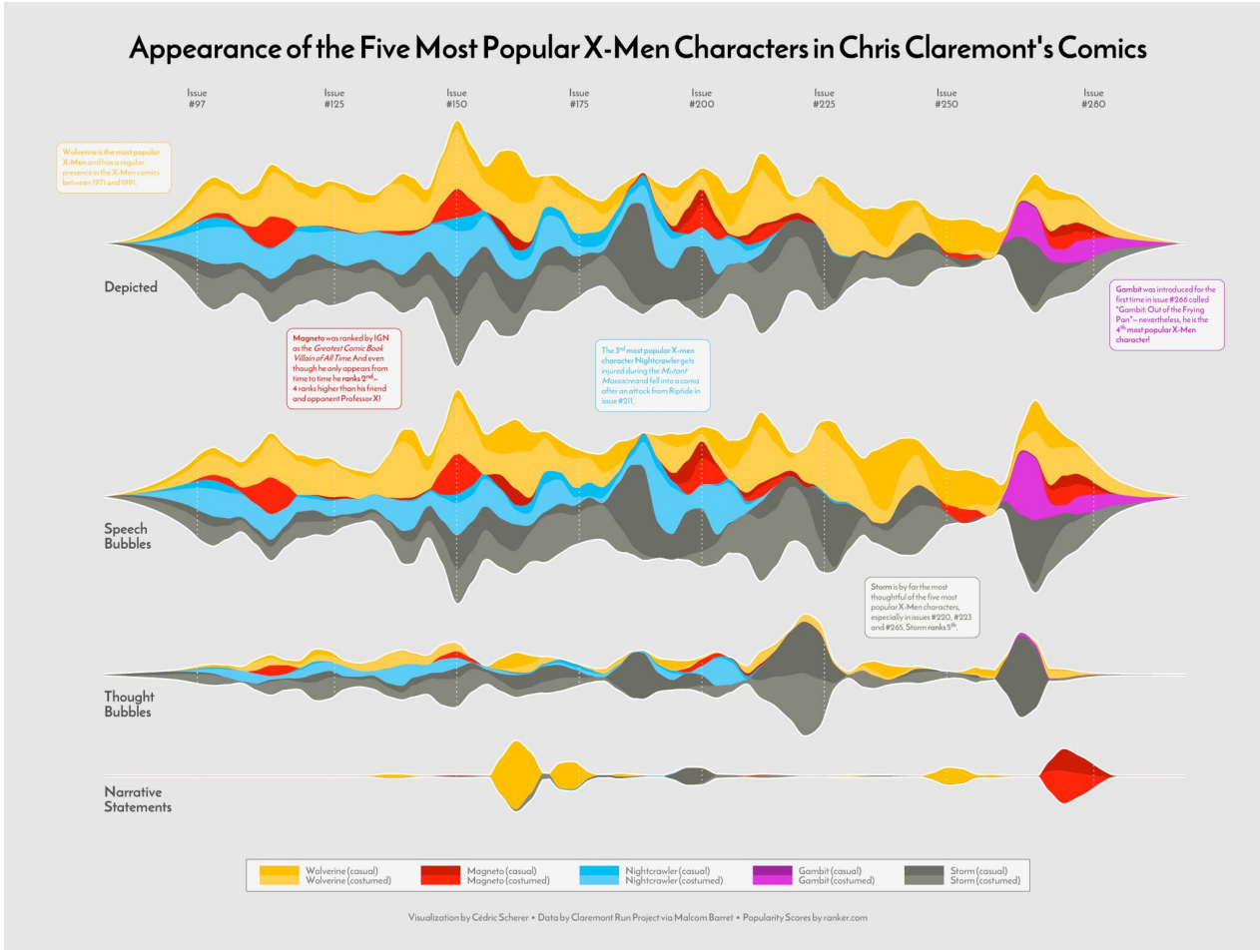
Remember this from Week 2?

Idiom: Dot Plot

- What?
  - **"Simple" dot plot**: One quantitative attributes per item (can have multiple items on one scale)
  - **Connected dot plot**: Two (or more quantitative attributes per item)
  - Zero or one categorical attribute per dot
  - Multiple items can be placed
- Why?
  - Find trends, outliers, ranges
- How?
  - Position
  - Scalability: from a few items to hundreds of items

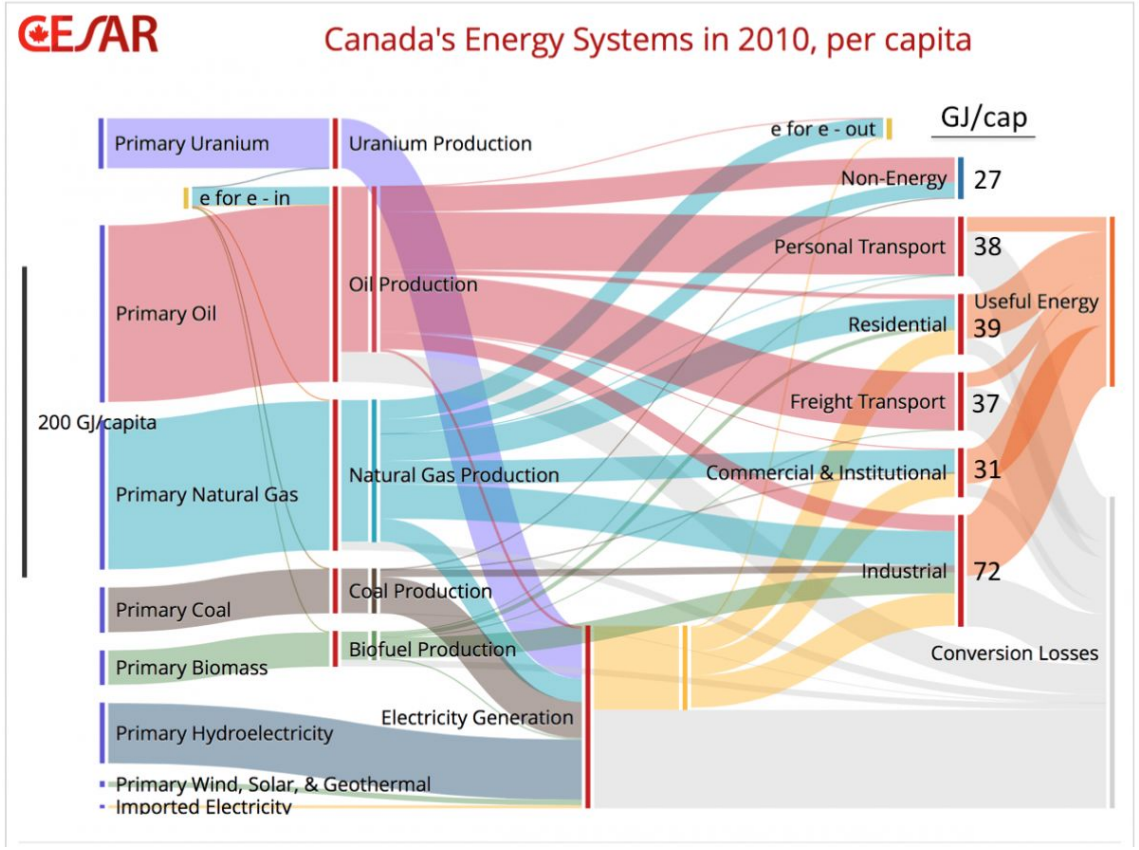
Activity 1 – Identifying Idioms (Group A)

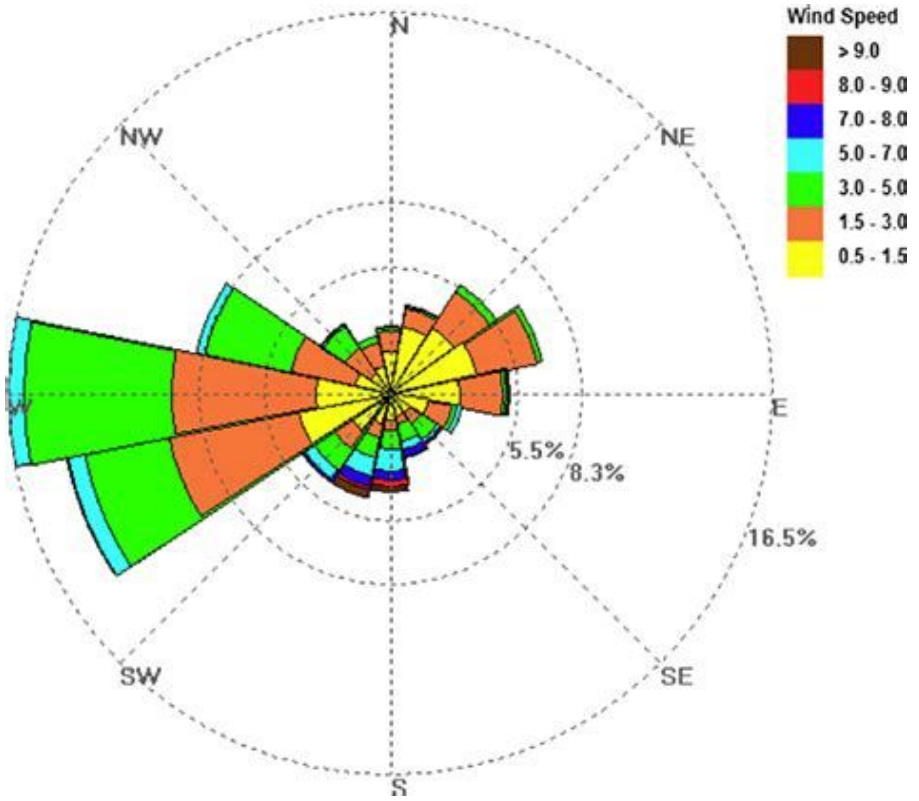




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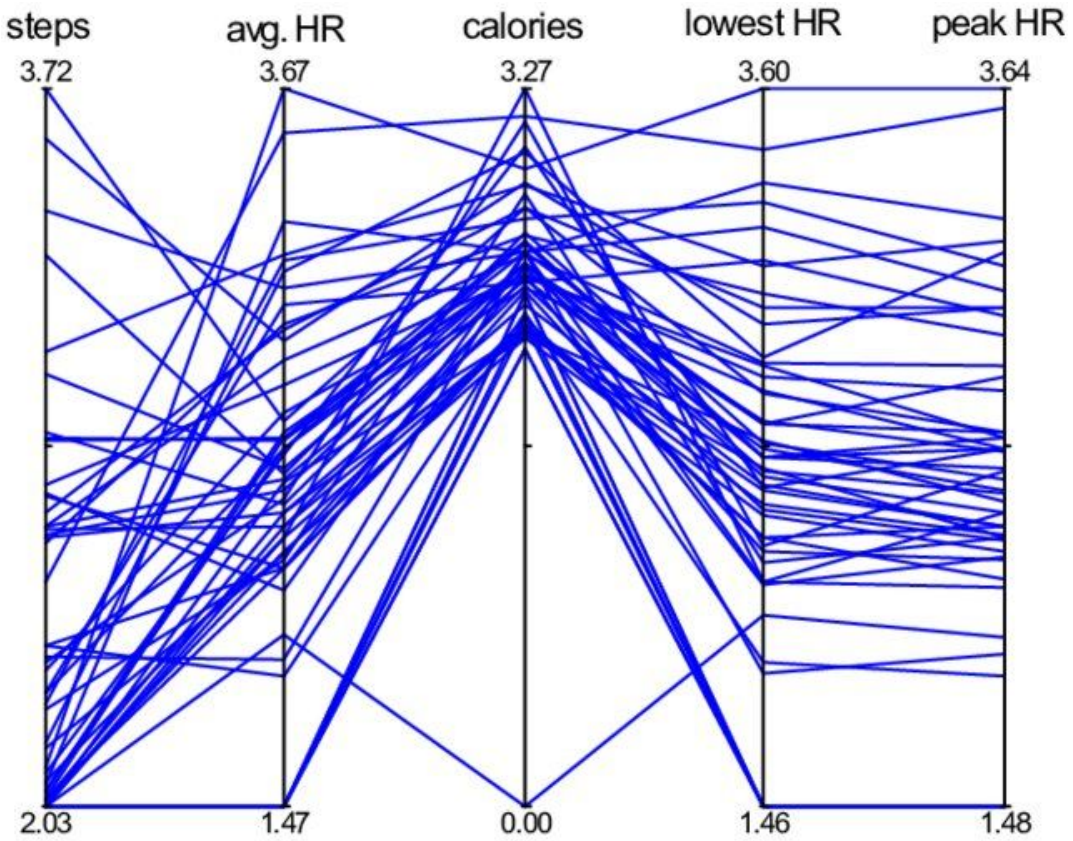
Activity 1 – Identifying Idioms (Group B)

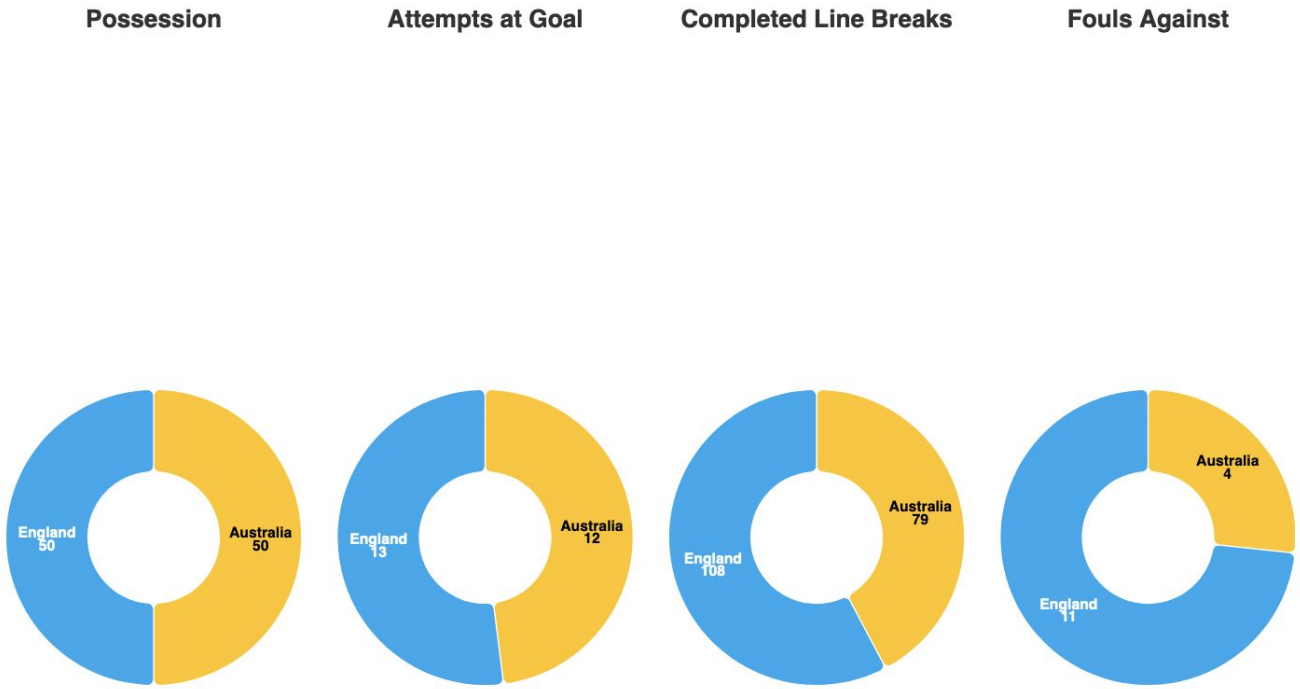




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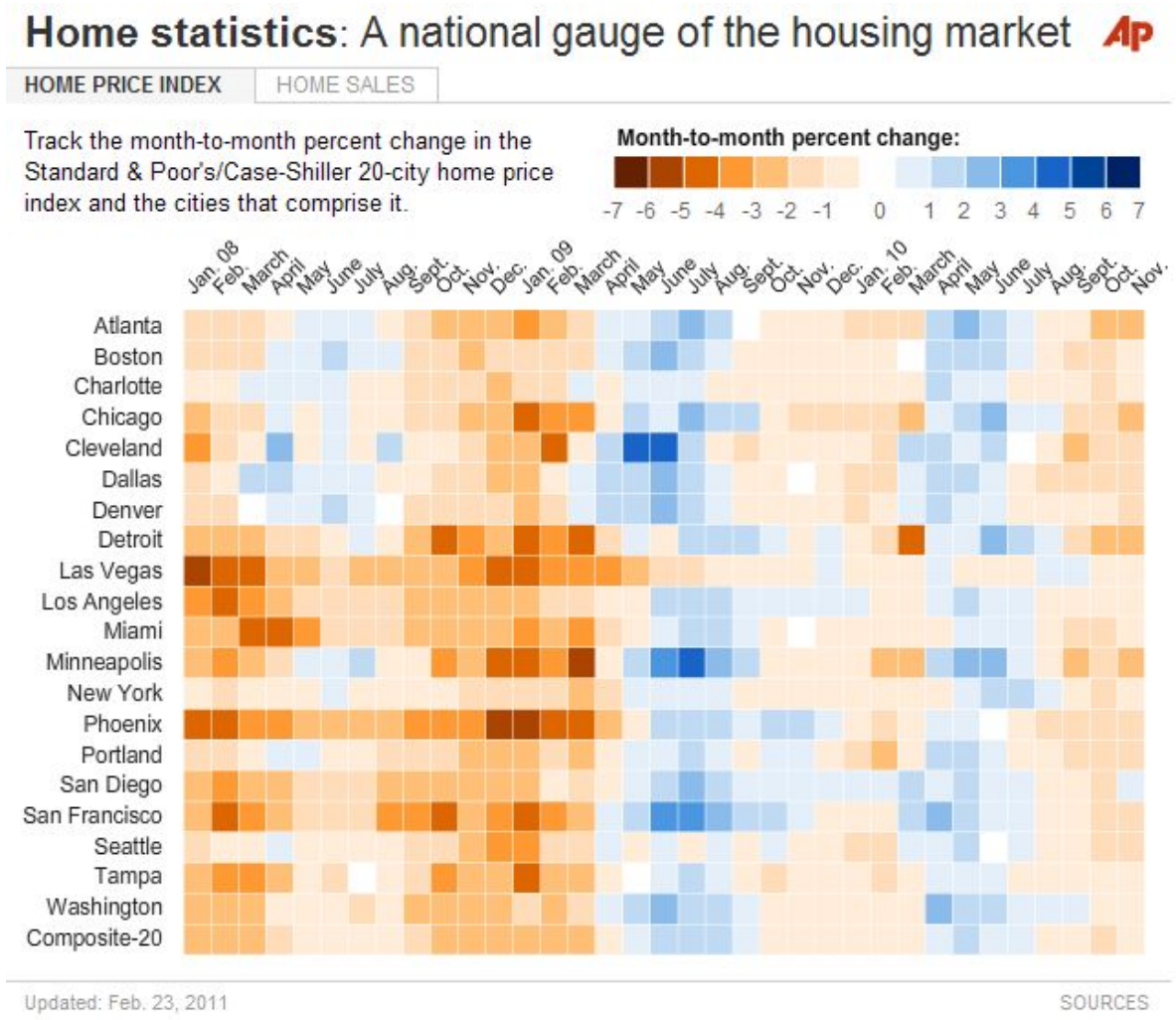
Activity 1 – Identifying Idioms (Group C)

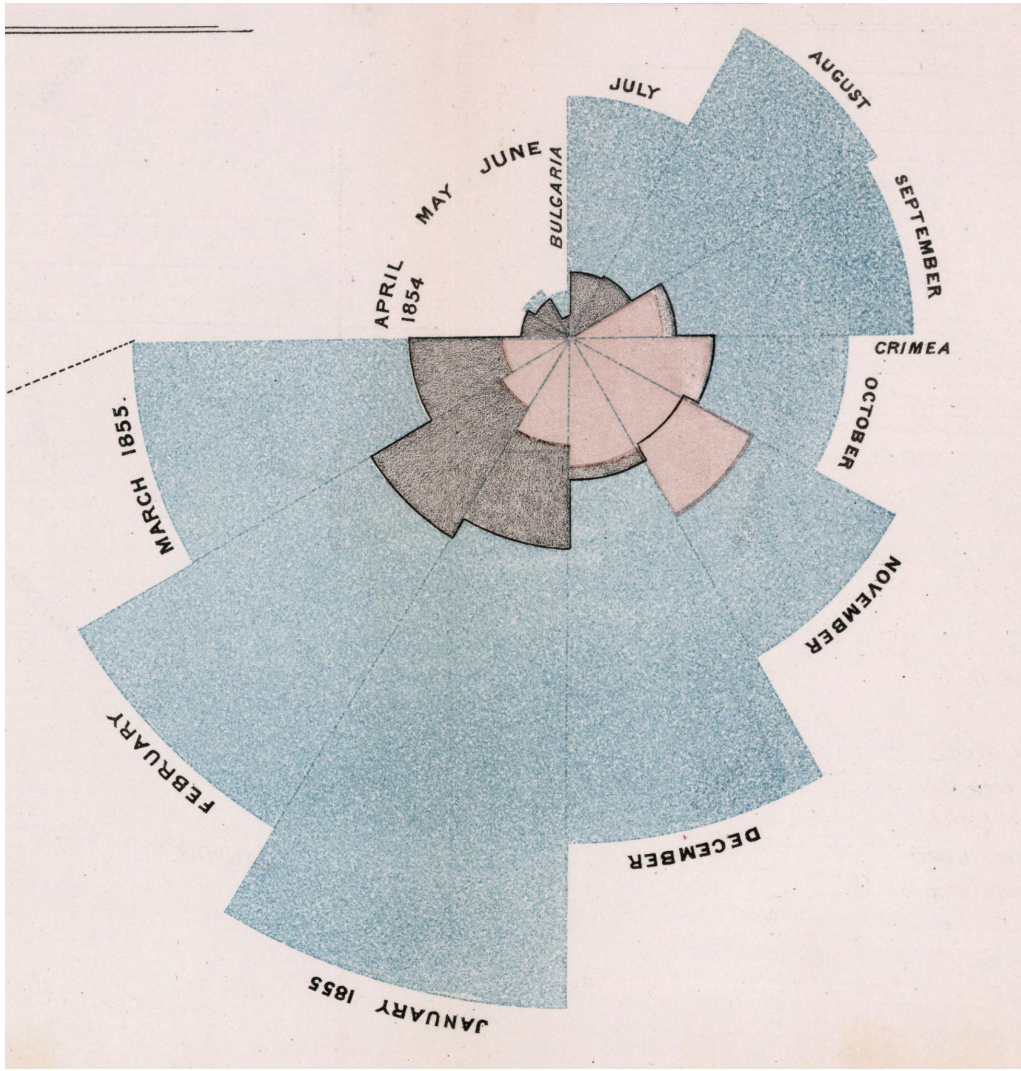




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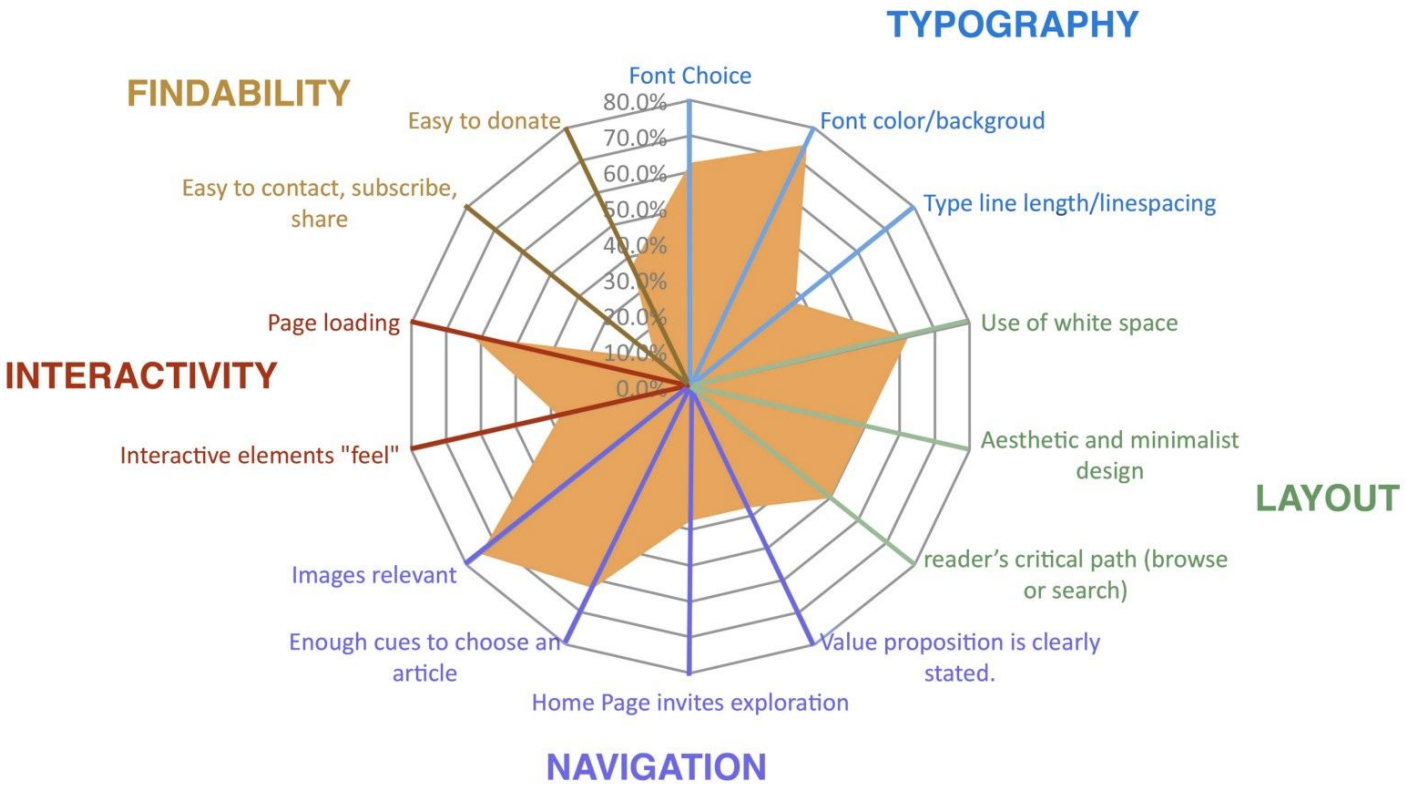
Activity 1 – Identifying Idioms (Group D)

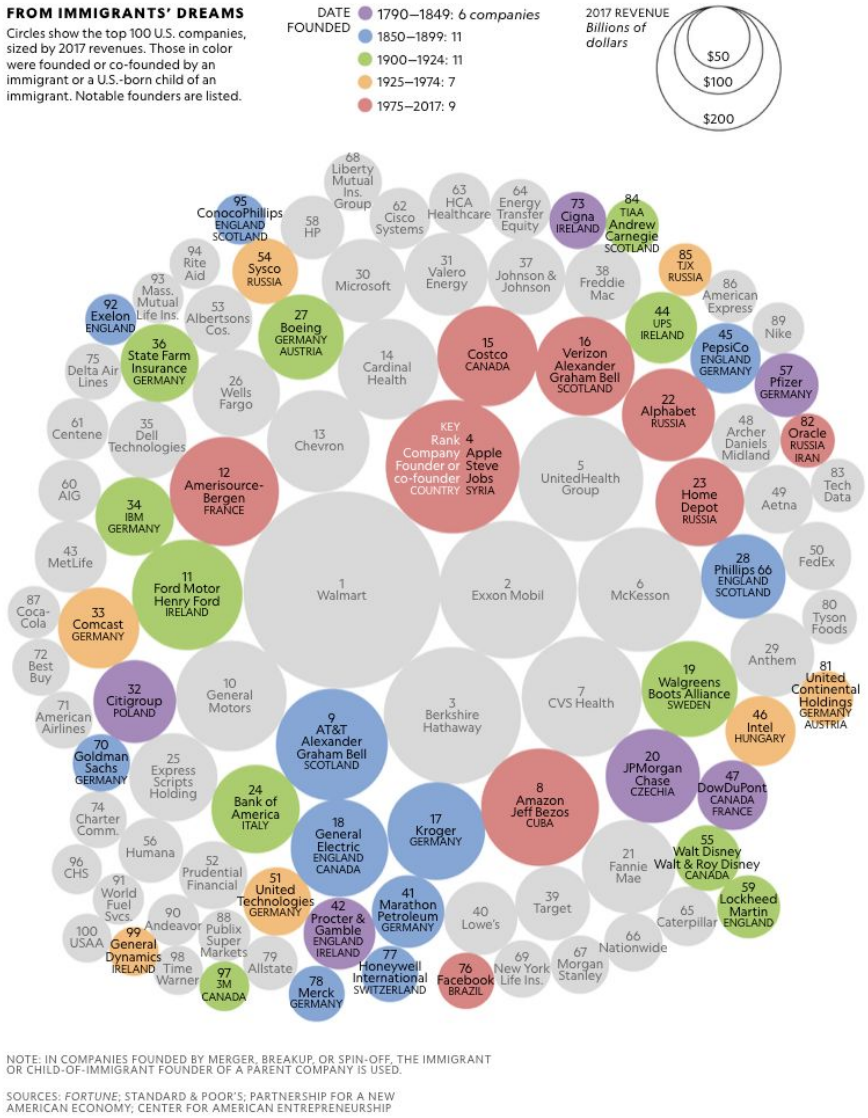




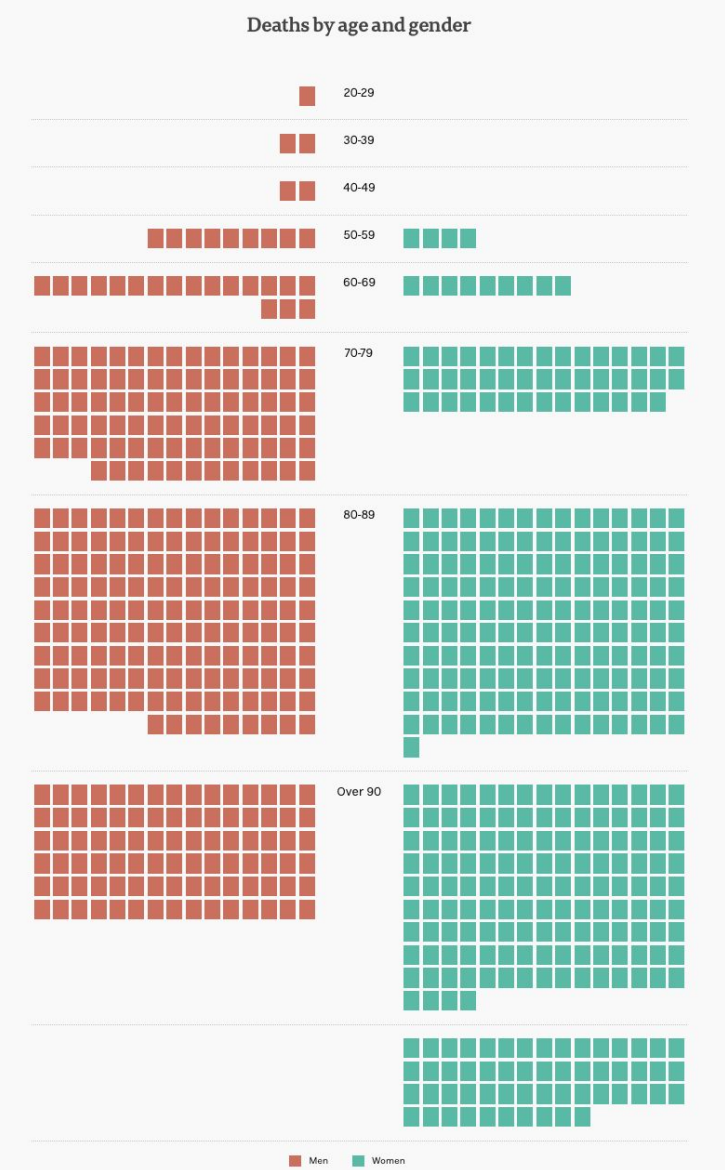
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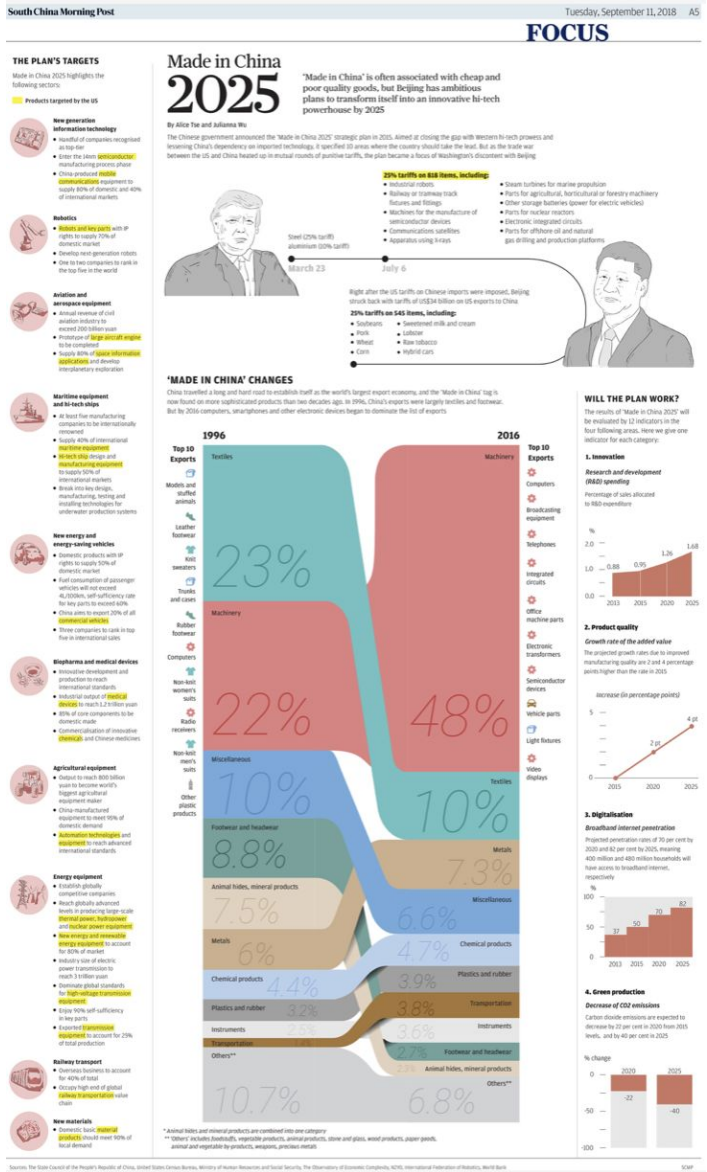
Activity 1 – Identifying Idioms (Group E)

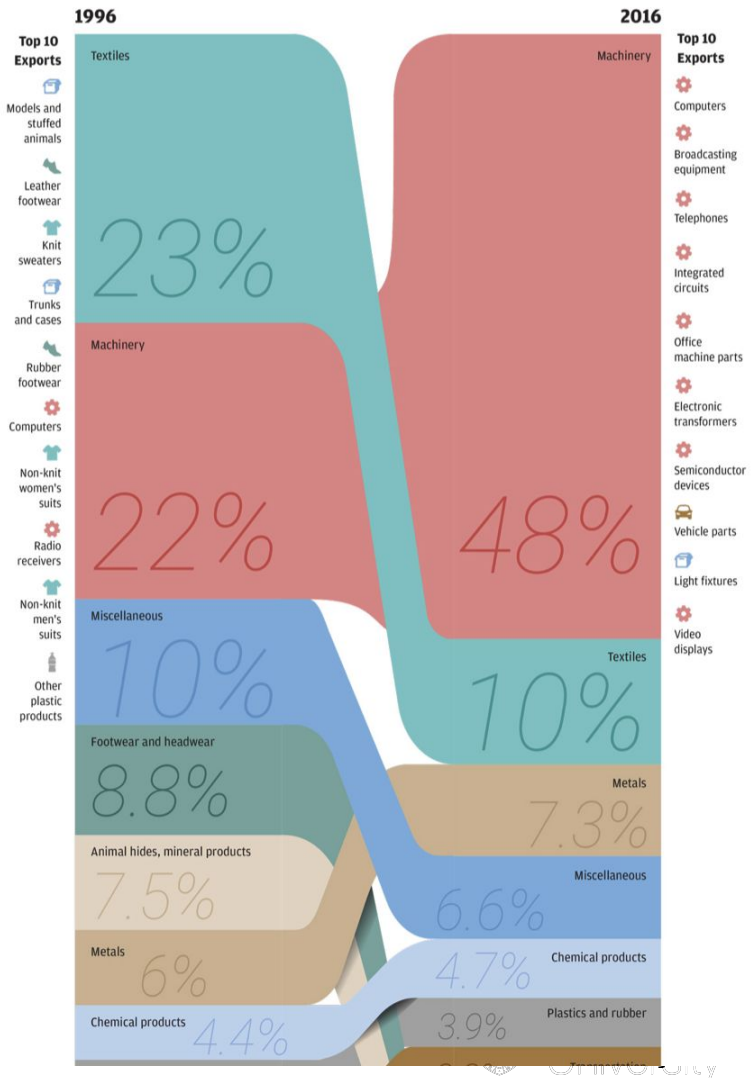




Activity 1 – Identifying Idioms (Group F)







Activity 2 – Ranking Idioms

For each of the idioms, consider whether you find it

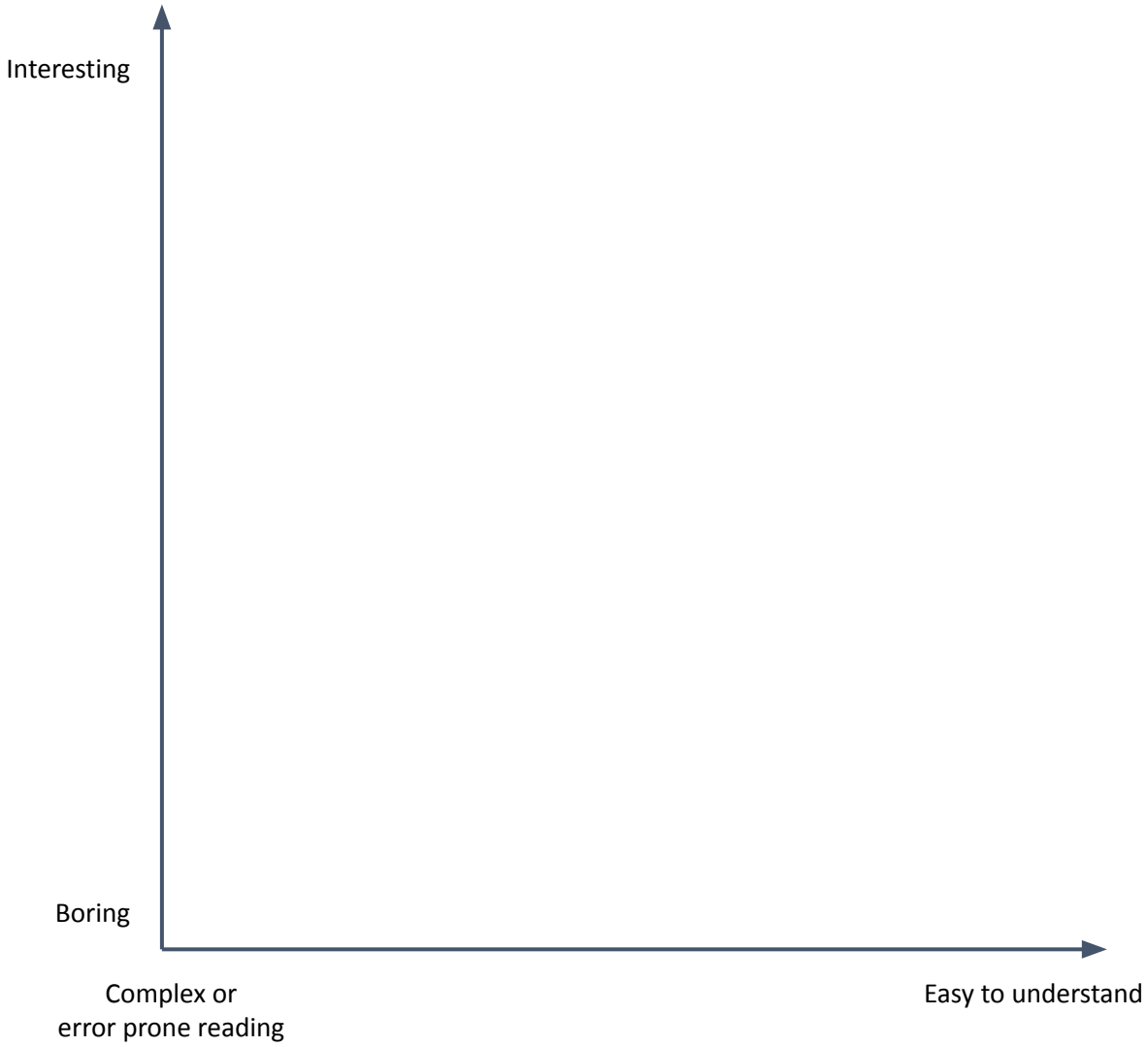
- *Interesting* or *Boring*
- *Complex / error-prone to read*
  or *Easy to understand*

Place the relevant icon on your graph in a suitable position.
Copy/duplicate the draft page if needed.

The icons are from [The Data Visualisation Catalogue]



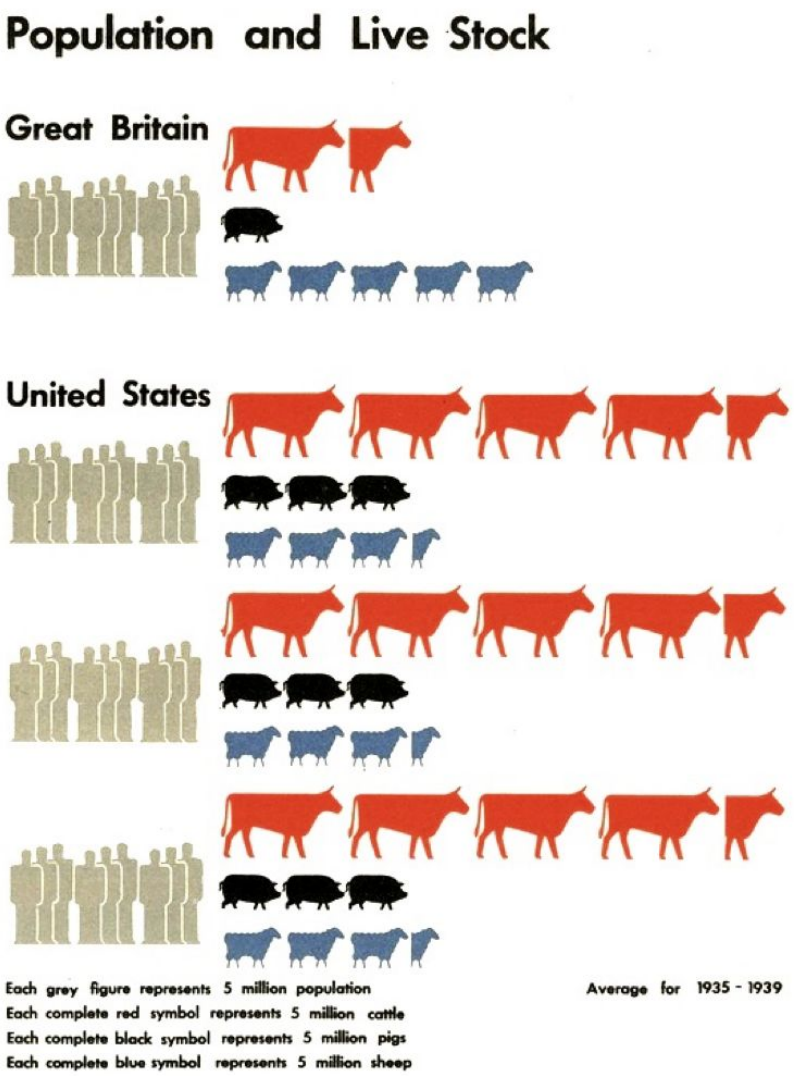
Activity 2: Ranking Idioms





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Activity 3: Isotype



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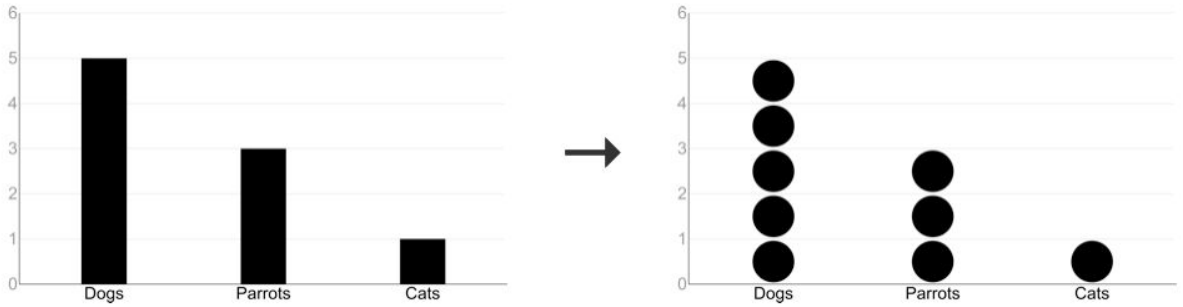
Activity 3: Isotype

1. Stacked icons allow you to encode data as both length and quantity

The human visual system relies on different approaches for judgments of length (like the height of a tree) and quantity (such as counting the fruit on the branches).

Chart-making tip: Splitting a continuous bar into a stack of items allows the visual system to code both length (the height of the stack) and quantity (the number in the stack). We found that this stacking benefits immediate recall for the data in a briefly presented graph. Because the visual system is incredibly efficient at quantifying groups of 1-5 objects, we recommend splitting stacks into no more than 4-5 segments – we found no additional benefit for splitting for larger collections.

Caveat: This benefit only held for short-term recall of data, not long-term recall, and there was no evidence that it sped up data extraction. Critically, we found no evidence for impairments due to splitting – at the very least, it doesn't hurt.



http://steveharoz.com/research/isotype/

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Activity 3: Isotype

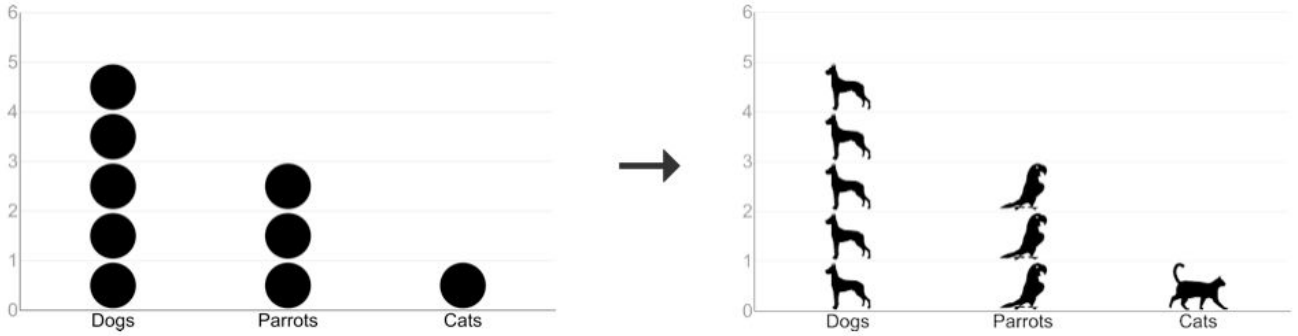
2. Icons aid memory and are more engaging

Long-term memory is aided by additional information that 'hooks' to what you want to remember – you'll better recall what you had for breakfast yesterday after someone reminds you of the name of the restaurant. Using real visual objects to encode data seems to lead to a mild improvement in remembering that data – you could better recall that daily production was 5 cars if you can recall that there was a stack of 5 cars.

Chart-making tip: Using pictographic icons can help with long-term memory of information.

Caveat: The benefit to long-term memory is small, and there's no benefit to speed or working memory.

Chart-making tip: We also found that ISOTYPE style charts were more engaging – when given a choice to view graphs of that style vs. standard bar graphs, our viewers had a clear initial preference for ISOTYPE.



http://steveharoz.com/research/isotype/

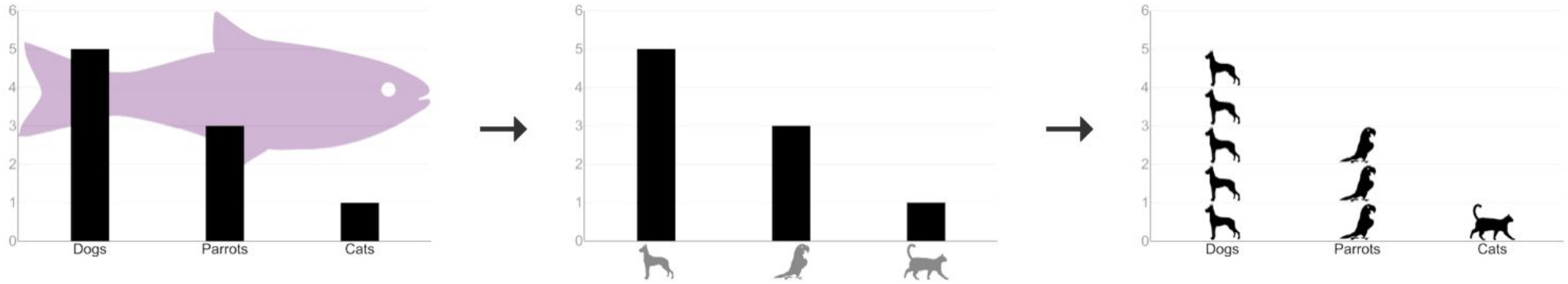
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Activity 3: Isotype

3. Images that are not used to depict data are a distraction

We added images to the background and labels, which did not depict data. Short-term memory retention for the data dropped by 40%, and the speed of finding information dropped dramatically.

Chart-making tip: Images can be used to depict data, but background images or pictographic labels could merely distract.

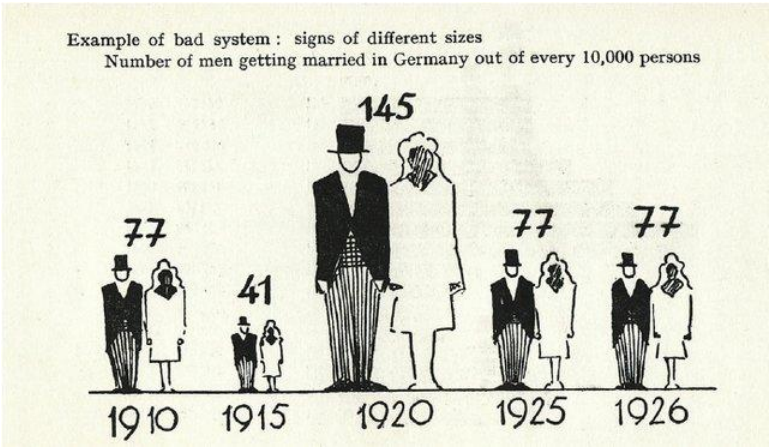


http://steveharoz.com/research/isotype/

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Activity 3: Isotype





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Activity 3: Isotype

Task 1: Sketch an Isotype visualisation that shows the number of dead, wounded and unhurt soldiers after World War I for the two opposing coalitions.

	Dead	Wounded	Unhurt
Allies: USA, UK, France, Belgium, Italy, Serbia, Romania, Russia, Japan	5.5 million	12 million	25.5 million
Central Powers: Germany, Austria Hungary, Bulgaria, Ottoman Empire	4 million	8.5 million	12.5 million

Numbers are estimated. Source: Wikipedia

Task 2: Design an Isotype visualisation that shows the chances of survival of children with diphtheria depending on the delay between diagnosis and treatment.

	1 Day	2 Days	3 Days	4 Days	5 Days
Deaths	0%	5%	10%	15%	20%

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